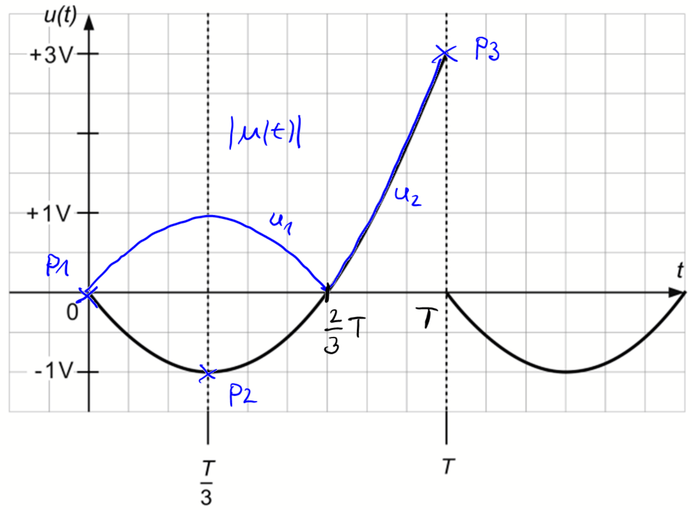
Übungsaufgabe PER2



$\overline{u} = 0\,V$

$\overline{|u|} = 0{,}89\,V$

$U_{RMS} = 1{,}095\,V$

a) Parabelfunktion allgemein:

$$u(t) = a t^2 + b t + c$$

$P1)$ $u(t=0) = 0\,V \Rightarrow a\,0^2 + b\,0 + c = 0\,V \Rightarrow c = 0$

$P2)$ $u(t=\frac{1}{3}T) = -1\,V \Rightarrow a \cdot \left(\frac{1}{3}T\right)^2 + b\left(\frac{1}{3}T\right) = -1\,V \quad (2)$

$P3)$ $u(t=T) = 3\,V \Rightarrow a\,T^2 + b\,T = 3\,V \quad (3)$

$(3) \Rightarrow b\,T = 3\,V - a\,T^2$

$$b = \frac{3\,V}{T} - \frac{a\,T^2}{T} = \frac{3\,V}{T} - a\,T$$

b in $(2) \Rightarrow a \cdot \left(\frac{1}{3}T\right)^2 + \left(\frac{3V}{T} - aT\right)\left(\frac{1}{3}T\right) = -1\,V$

$$a \cdot \frac{T^2}{9} + 1\,V - \frac{1}{3}\,a\,T^2 = -1\,V$$

$$\frac{1}{9}\,a\,T^2 - \frac{1}{3}\,a\,T^2 = -2\,V$$

$$-\frac{2}{9}\,a\,T^2 = -2\,V$$

$$a = -2\,V \cdot \left(-\frac{9}{2\,T^2}\right)$$

$$= \frac{9\,V}{T^2}$$

$$b = \frac{3V}{T} - \frac{9V}{T^2}\,T = \frac{3V}{T} - \frac{9V}{T} = -\frac{6V}{T}$$

$$\Rightarrow \quad u(t) = \underbrace{\frac{9V}{T^2}}_{a} t^2 \underbrace{- \frac{6V}{T}}_{b} t + \underbrace{0}_{c}$$

b) Arithmetischer Mittelwert

$$\overline{u} = \frac{1}{T}\int_0^T u(t)\,dt$$

$$= \frac{1}{T}\int_0^T \frac{9V}{T^2}t^2 - \frac{6V}{T}t \quad dt$$

$$= \frac{1}{T}\left[\frac{9V}{T^2}\frac{t^3}{3} - \frac{6V}{T}\frac{t^2}{2}\right]_0^T$$

$$= \frac{1}{T}\left[\frac{9V}{\cancel{T^2}}\frac{T^{\cancel{3}}}{3} - \frac{6V}{\cancel{T}}\frac{T^{\cancel{2}}}{2}\right]$$

$$= 3V - 3V$$

$$= \underline{\underline{0V}}$$

c) Gleichrichtwert

$$\overline{|u|} = \frac{1}{T}\int_0^T |u(t)|\,dt = \frac{1}{T}\left[\int_0^{\frac{2}{3}T} |u_1(t)|\,dt + \int_{\frac{2}{3}T}^{T} |u_2(t)|\,dt\right]$$

$$= \frac{1}{T}\left[\int_0^{\frac{2}{3}T} \underbrace{\left(-\frac{9V}{T^2}t^2 + \frac{6V}{T}t\right)}_{-u(t)} dt + \int_{\frac{2}{3}T}^{T} \underbrace{\left(\frac{9V}{T^2}t^2 - \frac{6V}{T}t\right)}_{u(t)} dt\right]$$

$$= \frac{1}{T}\left[\left(-\frac{9V}{T^2}\frac{t^3}{3} + \frac{6V}{T}\frac{t^2}{2}\right)\Bigg|_0^{\frac{2}{3}T} + \left(\frac{9V}{T^2}\frac{t^3}{3} - \frac{6V}{T}\frac{t^2}{2}\right)\Bigg|_{\frac{2}{3}T}^{T}\right]$$

$$= \frac{1}{T}\Big[-\frac{9V}{T^2}\left(\frac{2}{3}T\right)^3\frac{1}{3} + \frac{6V}{T}\left(\frac{2}{3}T\right)^2\frac{1}{2} + \frac{9V}{T^2}\frac{T^3}{3} - \frac{6V}{T}\frac{T^2}{2}$$

$$-\frac{9V}{T^2}\left(\frac{2}{3}T\right)^3\frac{1}{3} + \frac{6V}{T}\left(\frac{2}{3}T\right)^2\frac{1}{2}\Big]$$

$$= \frac{1}{T}\Big[-\frac{9V}{\cancel{T^2}}\frac{8}{27}\frac{1}{3}T^{\cancel{3}} + \frac{6V}{\cancel{T}}\frac{4}{9\cdot 2}T^{\cancel{2}} + \frac{9V}{\cancel{T^2}}\frac{T^{\cancel{3}}}{3} - \frac{6V}{\cancel{T}}\frac{T^{\cancel{2}}}{2}$$

$$-\frac{9V}{\cancel{T^2}}\frac{8}{27}\frac{1}{3}T^{\cancel{3}} + \frac{6V}{\cancel{T}}\frac{4}{9}\frac{T^{\cancel{2}}}{2}\Big]$$

$$= -\frac{3V\cdot 8}{\cancel{27}\,9} + \frac{3\cdot 4V}{9} + \cancel{3V} - \cancel{3V} - \frac{3V\cdot 8}{\cancel{27}\,9} + \frac{3V\cdot 4}{9}$$

$$= -\frac{8}{9}\cdot 2V + \frac{4}{3}V\cdot 2$$

$$= -\frac{16}{9}V + \frac{8}{3}V = -\frac{16}{9}V + \frac{24}{9}V$$

$$= +\frac{8}{9}V \approx 0{,}89V$$

d)

$$U_{RMS} = \sqrt{\frac{1}{T}\int_0^T u^2(t)\,dt}$$

$$= \sqrt{\frac{1}{T}\int_0^T \left(\frac{9V}{T^2}t^2 - \frac{6V}{T}t\right)^2 dt}$$

$(a-b)^2 = a^2 - 2ab + b^2$

$$= \sqrt{\frac{1}{T}\int_0^T \left(\frac{9V}{T^2}t^2\right)^2 - 2\left(\frac{9V}{T^2}\right)\left(\frac{6V}{T}\right)t^3 + \left(\frac{6V}{T}t\right)^2 dt}$$

$$= \sqrt{\frac{1}{T}\left[\left(\frac{9V}{T^2}\right)^2\frac{t^5}{5} - \frac{108V^2}{T^3}\frac{t^4}{4} + \left(\frac{6V}{T}\right)^2\frac{t^3}{3}\right]\Bigg|_0^T}$$

$$= \sqrt{\frac{1}{T}\left[\frac{81V^2}{\cancel{T^4}}\frac{T^{\cancel{5}}}{5} - \frac{108V^2}{\cancel{T^3}}\frac{T^{\cancel{4}}}{4} + \frac{36V^2}{\cancel{T^2}}\frac{T^{\cancel{3}}}{3}\right]}$$

$$= \underline{\underline{1{,}095V}}$$